

STARVNT EVENT OS

AURA+ — FULL CUSTOMER + VENDOR WEBSITE & EVENT/VENDOR UNIVERSE

Developer Master Blueprint • Product Specification • UX • Architecture • Data • API • Matching • Execution

Customer designs the experience. Vendor executes the experience. StarVnt Core orchestrates everything.

Purpose: give product, frontend, backend, AI, DevOps and admin developers one shared specification for building the StarVnt Event OS without reducing Aura+ to a simple chatbot or fixed wedding form.

1. EXECUTIVE PRODUCT DEFINITION

StarVnt Event OS is an AI-powered event operating system. Customers can design any supported event through conversation, voice, visuals, reference images and direct controls. Aura+ converts those decisions into structured executable requirements. StarVnt Core calculates cost, checks feasibility, matches real-time vendor capacity, validates scope, applies commercial rules, manages booking/payment and tracks execution through an Event Digital Twin.

Aura+ has two role-specific experiences:

• Customer Aura+: understands the desired experience, visualizes it, calculates the customer-facing budget and guides design.

• Vendor Aura+: interprets the execution brief, shows reference + AI breakdown, checks scope, assists quotation and supports execution.

• StarVnt Core: authoritative operational layer for state, cost, matching, commercial isolation, assignment, booking and execution.

Core product promise

Customer: “I can create my event my way.”

Vendor: “I know exactly what I need to execute.”

StarVnt: “I orchestrate the event from idea to execution.”

2. NON-NEGOTIABLE PRODUCT PRINCIPLES

1. One Question → One Decision. Aura+ never dumps a long questionnaire when one next decision is enough.

2. Never ask a question just to collect data. Ask only when the answer can change the event, price, feasibility, experience or execution.

3. Aura+ guides the customer but never traps the customer. Customer can add, edit, remove, replace, upload and describe ideas directly.

4. Customer design and vendor operations run in parallel. The customer must not wait for vendor responses.

5. 60 seconds is a vendor response SLA, not a vendor-selection SLA. Fastest vendor does not automatically win.

6. AI concept is never presented as guaranteed availability. Visuals carry source/status badges.

7. Event State is the single source of truth for customer actions, AI inference, visual state, budget, service requirements and downstream orchestration.

8. LLM is not the sole operational authority. Core/rules/data engines remain authoritative for significant decisions.

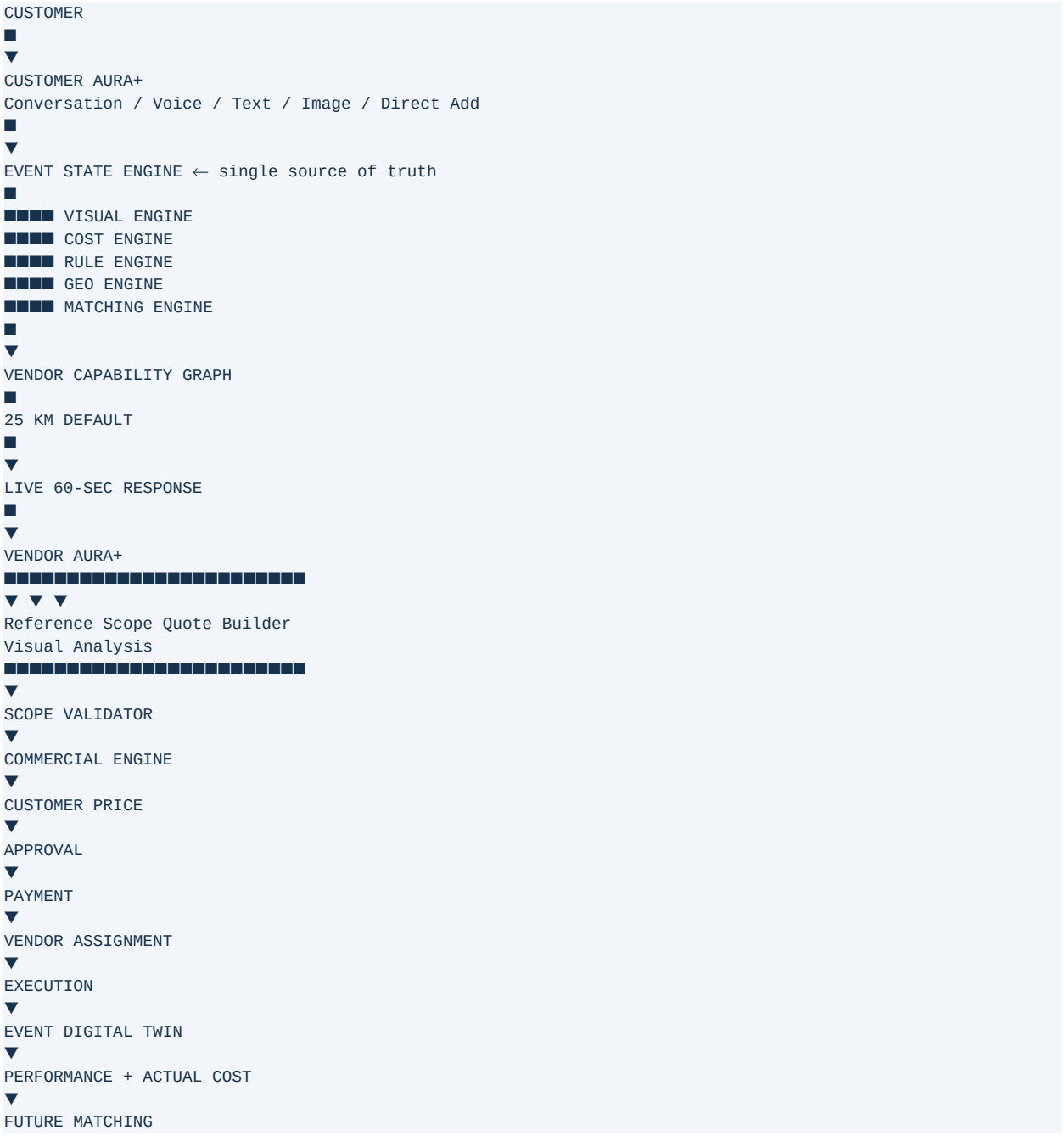
9. Customer and vendor economics are isolated at the API/data layer. UI hiding is not sufficient.

10. Never show an unavailable service as available.

11. Never search vendors before a requirement is sufficiently actionable.

12. Never make the customer wait while StarVnt operates in the background.

3. MASTER SYSTEM ARCHITECTURE



Three product surfaces

- Customer Website / App: design and manage the event.
- Vendor Website / App: receive opportunities, quote and execute.
- StarVnt Command Center: operate, monitor, validate, intervene and analyze.

4. CUSTOMER WEBSITE BLUEPRINT

Route	Screen	Developer purpose
/	Landing	Value proposition, Start Designing with Aura+, Explore Event Ideas.
/login	Authentication	Mobile OTP, Google, Email.
/dashboard	Customer Dashboard	My Events, readiness, budget, continue Aura+.
/event/:eventId/aura	Aura+ Studio	Primary conversational + visual workspace.
/event/:eventId/canvas	Live Event Canvas	Visual event state, services, budget and direct controls.
/event/:eventId/services	Service Catalogue	Contextual service catalogue filtered by event type.
/event/:eventId/reference	Visual Reference	Upload and analyze customer references.
/event/:eventId/budget	Budget	Live cost, categories, remaining budget and what-if scenarios.
/event/:eventId/availability	Availability	Validated execution capacity, not raw vendor competition.
/event/:eventId/review	Review	Final event configuration and customer price.
/event/:eventId/payment	Payment	Advance/payment state.
/event/:eventId	Event Tracking	Booking, execution milestones, readiness and risks.

Desktop Aura+ Studio



Mobile: event summary → live visual → conversation → persistent “ Add Something” → sticky budget.

5. CUSTOMER AURA+ — CONVERSATIONAL EVENT STUDIO

Aura+ is not a chatbot. It is a live AI Event Designer connected to the Event State Engine. Every answer, selection, uploaded image or direct edit becomes structured state.

Interaction model

Ask → Understand → Show → Calculate → Give control → Continue

Customer inputs

• Text: “White floral stage □□□.”

• Voice: natural spoken event requirements.

• Tap: visual cards, Add, Edit, Remove, Replace.

• Image: customer reference image(s).

• Open idea: “I want a big circular floral arch with warm lights.”

• Surprise Me: Aura+ proposes a balanced configuration based on event state and budget.

Persistent Add menu

```
■ Add Something
■■■ Search a Service
■■■ Choose a Design
■■■ Upload Reference
■■■ Describe My Idea
■■■ Surprise Me
```

Customer direct-control rule

Customer actions and Aura+ actions must enter the same Event State Engine. No random frontend-only state is allowed for commercially meaningful changes.

6. VISUAL ENGINE

Visual is not decoration on top of Aura+. Visual is a representation of Event State. When the state changes, the visual, price and execution requirement can change together.

```
VISUAL ENGINE
■
■■■ STARVNT PORTFOLIO
■ Actual executed work
■
■■■ STARVNT CATALOGUE
■ Approved/bookable design library
■
■■■ AI CONCEPT
Visualization / inspiration only
```

Mandatory visual badges

• STARVNT PORTFOLIO — actual/approved executed work.

• STARVNT CATALOGUE — approved design/service reference.

• AI CONCEPT — generated/inspiration visual; not an availability promise.

• CUSTOMER REFERENCE — customer-uploaded reference.

• VENDOR PROPOSAL — vendor-uploaded implementation reference.

Reference pipeline

```
Customer Image
↓
Vision Analysis
↓
Design Attributes
↓
Event Service Configuration
↓
Catalogue / Portfolio / Vendor Capability Search
↓
Executable Options
```

Example attributes

```
style: elegant
primaryColor: white
secondaryColor: gold
floralDensity: high
lighting: warm
stageSize: 10x20
material: fresh/artificial TBD
```

7. LIVE EVENT CANVAS + CUSTOMER CONTROL

Example UI

```
YOUR WEDDING
500 Guests · Kolkata

[ LIVE EVENT VISUAL ]

White + Gold · Elegant Luxury

SERVICES
✓ Venue
✓ Catering
✓ Stage
✓ Entrance
■ Photography
■ Entertainment

ESTIMATED
■4,84,000 / ■5,00,000
Remaining ■16,000

■ Add Something
■ Ask Aura+
```

Supported actions

• Add a service.

• Edit configuration.

• Remove a service.

• Replace a design.

• Upload/replace a reference.

• Describe a new idea.

• Ask Aura+ to create a what-if scenario.

• Ask Aura+ to suggest or surprise.

Validation examples

• If a customer asks for a 200-person venue for 500 guests, Core rejects the invalid configuration and Aura+ explains the constraint.

• If a premium service cannot realistically fit the requested budget, Aura+ presents alternatives or a budget adjustment rather than pretending it is feasible.

• If a service is not relevant to the event type, it should not be surfaced by default.

8. COST, BUDGET & COMMERCIAL ENGINE

Customer-facing budget is not vendor cost. The system must maintain separate commercial layers.

```
VendorQuotation
↓
StarVntCommercial
↓
CustomerPrice

Example:
Vendor Cost ■45,000
Internal Target ■52,000
Customer Price ■56,000

Customer API → customerPrice only
Vendor API → approved vendor amount/target only
Admin → full commercial view
```

Budget states

- WITHIN_BUDGET
- BUDGET_TIGHT
- OVER_BUDGET
- PRICE_PENDING
- AVAILABILITY_PENDING

What-if engine

```
Customer: "■50,000 ■■■■■ ■■■■"

Aura+ generates scenario objects:
A: Reduce Decoration → estimated saving
B: Simplify Photography → estimated saving
C: Remove selected service → estimated saving
D: Combination → estimated saving

Customer chooses; system applies the selected scenario to Event State.
```

9. REAL-TIME VENDOR CAPACITY MATCHING

Vendor search begins when a service requirement is sufficiently actionable. The full event does not need to be complete.

```
Known Requirement
↓
Cost Model
↓
Geo / Radius Rules
↓
Capability + Availability
↓
Eligible Vendor Pool
↓
Live 60-sec Response
↓
Scope Validation
↓
Core Scoring
↓
1–3 Validated Options
```

Default service radius rules

Service	Default radius
Decorator	25 km
Caterer	25 km
Photographer	30 km
Makeup Artist	20 km
DJ / Sound	30 km
Venue	20–25 km
Specialist Artist	50–100 km

Search waves

```
Wave 1: 0–25 km → 60 sec
No sufficient response?
Wave 2: 25–50 km → 60 sec
Still insufficient?
Wave 3: service-specific specialist range
or ask customer whether to expand / modify / continue without service.
```

10. VENDOR WEBSITE BLUEPRINT

Route	Screen	Purpose
/vendor	Vendor Landing	Value proposition and network onboarding.
/vendor/login	Vendor Auth	Mobile OTP / Google / Email.
/vendor/onboarding	Onboarding	Business → services → capability → area → pricing → portfolio → availability → verification.
/vendor/dashboard	Dashboard	Opportunities, active jobs, earnings and alerts.
/vendor/aura	Vendor Aura+	AI assistant for opportunities, scope and quotes.
/vendor/opportunities	Opportunities	Personalized relevant work feed.
/vendor/opportunities/:id	Opportunity Detail	Reference, AI analysis, execution blueprint, target range and response timer.
/vendor/jobs	Active Jobs	Assigned jobs and status.
/vendor/jobs/:id	Job Builder	Scope configuration and job state.
/vendor/jobs/:id/quote	Quote Builder	Vendor quote, inclusions, exclusions and validity.
/vendor/jobs/:id/execution	Execution	Milestones, proof and completion.
/vendor/calendar	Calendar	Availability and booking states.
/vendor/portfolio	Portfolio	Images, styles and executed work.
/vendor/services	Services	Capabilities and pricing.
/vendor/earnings	Earnings	Gross, paid, pending.
/vendor/performance	Performance	Response, reliability, scope accuracy and execution metrics.
/vendor/profile	Profile	Business identity and settings.

11. VENDOR AURA+ + OPPORTUNITY FLOW

Vendor Aura+ is Job Interpreter + Visual Brief + Quote Assistant + Execution Assistant.

```
NEW STARVNT OPPORTUNITY

Premium White Floral Stage
15 Dec 2026 · Kolkata
500 Guests
Target ■38,000–■43,000
Response window 60 sec

CUSTOMER REFERENCE
[image]

AI VISUAL ANALYSIS
White + Gold
Elegant
High Floral
Warm Lighting

EXECUTION BLUEPRINT
10×20 ft
Floral Backdrop
Warm Lighting
Setup
Dismantling

[AVAILABLE] [NOT AVAILABLE]
```

Vendor must not receive the customer's original budget if the commercial model requires budget isolation. Vendor sees StarVnt target range / job economics allowed for that vendor interaction.

Vendor response flow

```
Available
↓
Can you execute the complete scope?
↓
Yes / Need Clarification
↓
Estimated amount?
↓
Draft quotation
↓
Review
↓
Submit
```

Vendor can upload an actual implementation reference. StarVnt Core can use it as proposal evidence during scope validation.

12. SCOPE VALIDATION + EXECUTION

Customer changes must propagate through the same Event State. If a quoted scope changes, vendor receives a structured scope-change event.

```
Customer changes:
Gold accents: Standard → High

EVENT STATE UPDATED
↓
SERVICE CONFIG UPDATED
↓
COST RECALCULATED
↓
VENDOR REQUIREMENT UPDATED
↓
Vendor Aura+
[Accept] [Requote] [Cannot Execute]
```

Execution state machine

```
ASSIGNED
↓
ACCEPTED
↓
VENDOR_ARRIVED
↓
SETUP_STARTED
↓
STRUCTURE_COMPLETE
↓
DECORATION_COMPLETE
↓
QUALITY_CHECK
↓
READY
↓
COMPLETED
```

Proof types

- Arrival photo
- Setup photo
- Milestone photo
- Final result photo
- Completion proof
- Issue / exception proof

13. FULL EVENT UNIVERSE

Wedding & Marriage

Wedding • Reception • Engagement • Ring Ceremony • Haldi • Mehendi • Sangeet • Cocktail • Bachelor Party • Bachelorette • Wedding Anniversary • Pre-Wedding • Post-Wedding • Destination Wedding • Court Marriage Celebration • Bengali Wedding • Hindu Wedding • Muslim Wedding • Christian Wedding • Sikh Wedding • Jain Wedding • Interfaith Wedding • Traditional Wedding • Luxury Wedding • Intimate Wedding

Birthday & Personal

1st Birthday • Kids Birthday • Teen Birthday • Adult Birthday • Milestone Birthday • 18th • 21st • 25th • 30th • 40th • 50th • 60th • 70th+ • Surprise Birthday • Birthday Dinner • Birthday Brunch • Pool Party • Theme Party • Character Party • Gaming Party • Princess Party • Superhero Party

Couple & Relationship Events

Proposal • Engagement • Roka • Tilak • Couple Dinner • Couple Photoshoot • Pre-Wedding Shoot • Save-the-Date • Anniversary • Vow Renewal

Family & Social

Baby Shower • Godh Bharai • Annaprashan • Mundan • Rice Ceremony • Naming Ceremony • Housewarming • Griha Pravesh • Puja • Religious Gathering • Family Reunion • Farewell • Retirement • Reunion • Private Dinner • Family Celebration

Corporate

Conference • AGM • Town Hall • Seminar • Workshop • Training • Product Launch • Brand Launch • Dealer Meet • Distributor Meet • Sales Meet • Leadership Summit • Business Summit • Networking Event • Award Ceremony • Employee Engagement • Team Building • Annual Party • Corporate Dinner • Office Party • Festive Celebration • Foundation Day • Company Anniversary • Press Conference • Media Event • Investor Event • Roadshow • Exhibition • Trade Show • Expo • Recruitment Event • Campus Event • CSR Event • MICE

Entertainment

Concert • Live Music • DJ Night • Music Festival • Cultural Festival • Dance Show • Comedy Show • Stand-up Comedy • Fashion Show • Theatre • Award Show • Celebrity Appearance • Artist Meet & Greet • Open Mic • Poetry Event • Battle of Bands • College Fest • School Fest • Talent Show

Educational

School Annual Function • College Fest • Freshers • Farewell • Convocation • Seminar • Workshop • Conference • Hackathon • Science Fair • Career Fair • Educational Exhibition • Parent-Teacher Event • Alumni Meet • Orientation • Training Program

Religious / Cultural

Durga Puja • Kali Puja • Lakshmi Puja • Saraswati Puja • Ganesh Chaturthi • Diwali • Holi • Janmashtami • Christmas • Eid Gathering • Iftar • Religious Conference • Bhajan • Kirtan • Satsang • Cultural Programme • Community Festival

Public / Community

Public Meeting • Community Gathering • Awareness Programme • Public Conference • Rally-related Infrastructure • Press Interaction • Public Convention

Private / Lifestyle

Pool Party • Rooftop Party • House Party • Dinner Party • Cocktail Party • BBQ • Brunch • High Tea • Luxury Dinner • Yacht Party • Resort Party • Weekend Getaway • Friends Reunion

Exhibition / Trade

Exhibition • Trade Fair • Expo • Product Display • Automobile Expo • Real Estate Expo • Fashion Exhibition • Art Exhibition • Food Exhibition • Handicraft Exhibition • Startup Expo • B2B Exhibition

Destination

Destination Wedding • Destination Birthday • Corporate Retreat • Team Offsite • Resort Event • Beach Event • Hill Station Event • Heritage Event • International Event

Virtual / Hybrid

Webinar • Virtual Conference • Hybrid Conference • Virtual Product Launch • Online Workshop • Virtual Award Show • Live Streaming • Hybrid AGM • Online Training • Virtual Celebration

14. MASTER VENDOR UNIVERSE

Venue

Banquet Hall • Marriage Hall • Wedding Venue • Resort • Hotel • Rooftop • Lawn • Farmhouse • Community Hall • Club • Restaurant • Café • Convention Centre • Auditorium • Stadium • Open Ground • Beach Venue • Heritage Venue • Palace • Temple Venue • Church Venue • Private Villa • Warehouse • Industrial Venue • Corporate Office • Co-working Space • School • College • University • Exhibition Centre • Theatre • Studio • Photo Studio

Decoration

Stage Decoration • Mandap • Wedding Stage • Reception Stage • Haldi Decor • Mehendi Decor • Sangeet Decor • Engagement Decor • Ring Ceremony Decor • Birthday Decor • Corporate Stage • Exhibition Decor • Entrance • Gate • Arch • Floral Arch • Backdrop • Ceiling • Draping • Table Decor • Chair Decor • Centerpieces • Aisle Decor • Jaimala Stage • Photo Booth • Selfie Booth • Welcome Board • Signage • Couple Name Board • Neon Sign • LED Wall • Floral Tunnel • Hanging Decor

Florist

Fresh Flower • Artificial Flower • Imported Flower • Local Flower • Floral Installation • Bouquet • Garland • Bridal Bouquet • Centerpiece • Mandap Floral • Stage Floral • Entrance Floral • Hanging Floral • Flower Wall

Lighting

Ambient Lighting • Decorative Lighting • Fairy Lights • String Lights • LED • Moving Head • Beam • Par Can • Profile Light • Architectural Lighting • Stage Lighting • Wedding Lighting • Intelligent Lighting • Follow Spot • Uplighting • Chandeliers • Festoon • Laser • Special Effects Lighting

Sound / AV

Sound System • PA System • Line Array • Speakers • Subwoofer • Mixer • Microphone • Wireless Mic • Podium Mic • Conference Mic • Stage Monitor • LED Screen • Projection • Projector • Video Wall • LED Wall • Teleprompter • Confidence Monitor • Streaming • Recording

DJ / Music

DJ • DJ + Sound • Live Band • Singer • Instrumentalist • Guitarist • Pianist • Violinist • Tabla • Dhol • Dhol Tasha • Percussion • Orchestra • Folk Artist • Classical Artist • Bengali Band • Bollywood Band

Entertainment

Anchor • MC • Comedian • Magician • Mentalist • Dancer • Dance Group • Folk Performer • Fire Performer • Celebrity Artist • Mimicry Artist • Puppet Show • Kids Entertainer • Character Artist • Mascot

Photography

Candid Photography • Traditional Photography • Cinematic Photography • Editorial Photography • Documentary • Drone • Pre-Wedding • Engagement • Wedding • Reception • Corporate • Product • Event Photography

Videography

Traditional Video • Cinematic Film • Highlight Film • Full Event Film • Wedding Film • Same-Day Edit • Teaser • Reels • Social Content • Live Streaming • Multi-Camera • Drone Video

Makeup & Beauty

Bridal Makeup • Groom Makeup • Party Makeup • HD Makeup • Airbrush • Hair Styling • Saree Draping • Dupatta Draping • Groom Styling • Mehendi • Nail Artist • Beauty Artist

Bridal / Groom Fashion

Bridal Saree • Lehenga • Gown • Groom Sherwani • Suit • Tuxedo • Accessories • Jewellery • Bridal Jewellery • Groom Accessories • Rental Clothing

Catering

Bengali • North Indian • South Indian • Mughlai • Chinese • Continental • Italian • Thai • Mexican • Rajasthani • Punjabi • Gujarati • Jain • Vegan • International • Breakfast • Lunch • Dinner • Reception • Snacks • High Tea • Live Counter • Buffet • Plated • Cocktail • Dessert Counter • Beverage Counter

Special Food

Cake • Wedding Cake • Birthday Cake • Dessert • Mithai • Chocolate • Ice Cream • Live Chaat • Live Pasta • Live Pizza • Coffee • Tea • Mocktail • Beverage • Juice • Candy Floss • Popcorn

Hospitality

Hotel • Resort • Guest House • Homestay • Serviced Apartment • Hospitality Staff • Guest Relations • Concierge • Welcome Desk • Room Management

Transport

Sedan • SUV • Luxury Car • Vintage Car • Wedding Car • Limousine • Bus • Mini Bus • Tempo Traveller • Luxury Coach • Airport Transfer • Railway Transfer • Guest Shuttle • Driver Service • Valet Parking

Logistics

Event Logistics • Loading/Unloading • Transport • Storage • Equipment Transport • Furniture Rental • Tent • Barricade • Crowd Barrier • Generator • Power Backup • Electrical • Water • Waste Management

Furniture / Rental

Sofa • Chair • Table • Bar Table • Dining Table • Lounge Furniture • Stage Furniture • Couple Chair • Throne Chair • Royal Sofa • Carpet • Tent • Umbrella • Heater • Cooler • Air Cooler

Invitation / Print

Wedding Card • Digital Invitation • E-invite • WhatsApp Invite • RSVP • Welcome Board • Menu Card • Table Number • Signage • Brochure • Corporate Printing • Name Tags • Badges • Certificates

Gifting

Wedding Gift • Corporate Gift • Return Gift • Hamper • Personalized Gift • Leather Gift • Jewellery Gift • Chocolate Hamper • Premium Hamper • Employee Gift • VIP Gift

Invitation + RSVP

Digital Invite • RSVP Website • QR RSVP • Guest Tracking • WhatsApp RSVP • Reminder • Seating Management • Check-in

Event Staffing

Hostess • Promoter • Registration Staff • Ushers • Waiters • Bartenders where legally permitted • Housekeeping • Event Manager • Floor Manager • Runner • Guest Relations • Backstage Staff

Security

Security Guard • Bouncer • Female Security • VIP Security • Crowd Management • Entry Management • Metal Detector • Bag Check • Access Control

Medical / Safety

Ambulance • First Aid • Medical Staff • Doctor on Call • Fire Safety • Fire Extinguisher • Emergency Response

Special Effects

Cold Pyro where legally permitted • Cold Spark • Fog • Haze • CO2 Effects • Bubble Machine • Confetti • Streamers • Smoke Effects • Sparkular Effects • Projection Mapping

Event Technology

QR Check-in • RFID • Digital Registration • Event App • Live Poll • Digital Wall • Social Wall • LED • Projection • Live Streaming • Hybrid Event Platform • Event Website • Digital RSVP • Guest Management

Exhibition

Stall Fabrication • Exhibition Booth • Modular Stall • Custom Stall • Branding • Printing • LED • AV • Furniture • Registration Counter • Display Stand • Product Display • Lighting • Hostess • Logistics

Corporate Specialists

Conference Production • Stage Production • AV Production • Corporate Event Manager • Conference Moderator • MC • Speaker Management • Delegate Management • Registration • Travel Desk • Accommodation • Branding • Printing • Exhibition • Photography • Videography • Live Streaming

Artist / Celebrity

Singer • DJ • Band • Actor • Celebrity • Influencer • Comedian • Dancer • Performer • Speaker • Motivational Speaker • Celebrity Anchor

15. EVENT-SPECIFIC VENDOR GRAPHS

Wedding

```
Venue → Decoration → Stage / Mandap / Entrance / Floral / Lighting / Furniture → Food → Photography → Videography → Makeup → Mehendi → Bridal Fashion → Groom Fashion → Entertainment → Transport → Guest Stay → Invitation → Gifts → Logistics
```

Birthday

```
Venue → Theme Decor → Balloon Decor → Cake → Catering → DJ → Kids Entertainment → Character Artist → Photography → Videography → Return Gifts → Games → Magic Show → Invitation → Furniture → Lighting → Sound → Transport
```

Corporate Conference

```
Venue → Conference Production → Stage → AV → LED → Sound → Lighting → Branding → Printing → Registration → Hostess → MC → Speaker → Photography → Videography → Streaming → Catering → Transport → Accommodation → Security → Event Staffing
```

Concert

```
Venue → Stage → Truss → Sound → Line Array → Lighting → LED → Artist → DJ → Backline → Power → Generator → Security → Crowd Barrier → Ticketing → QR Entry → Medical → Photography → Videography → Streaming → Transport → Production Management
```

Exhibition

```
Venue → Stall Fabrication → Branding → Printing → LED → Lighting → Furniture → Registration → Hostess → AV → Internet → Security → Logistics → Transport → Photography → Videography
```

MICE / Retreat

```
Hotel / Resort → Travel → Airport Transfer → Bus → Rooms → Conference Hall → AV → Stage → Branding → Catering → Team Building → Entertainment → Photography → Videography → Gifts → Registration → Event Staff → Emergency Support
```

16. CORE DATA MODEL

Customer

id, name, phone, email, city, language, createdAt

Event

id, customerId, eventType, eventName, date, endDate, city, area, venueId, guestCount, budget, budgetFlexibility, status, readiness, createdAt, updatedAt

EventPreference

id, eventId, category, key, value, source, confidence

EventService

id, eventId, serviceType, serviceName, status, quantity, customerPrice, estimatedPrice, selectedAt

ServiceConfiguration

id, eventServiceId, style, colors, size, materials, requirements, referenceImages, specialInstructions

Vendor

id, businessName, ownerName, phone, email, city, serviceAreas, verificationStatus, rating, responseRate, reliabilityScore, createdAt

VendorService

id, vendorId, category, serviceName, description, basePrice, priceType, capacity, leadTime, active

VendorPricing

id, vendorServiceId, pricingModel, minPrice, maxPrice, guestMin, guestMax, location, season, validFrom, validTo

VendorAvailability

id, vendorId, date, startTime, endTime, status

VendorJob

id, eventId, eventServiceId, vendorId, scope, requirements, referenceImages, quotedPrice, status, assignedAt, deadline

VendorQuotation

id, vendorJobId, vendorId, subtotal, transport, labour, tax, total, validUntil, inclusions, exclusions, status, submittedAt

VendorProposal

id, vendorJobId, visualAssets, notes, claimedCapability, submittedAt

StarVntCommercial

id, eventServiceId, vendorCost, internalTarget, customerPrice, margin, contingency, ruleVersion

Booking

id, eventId, totalAmount, advanceAmount, remainingAmount, paymentStatus, bookingStatus, confirmedAt

Payment

id, bookingId, amount, gateway, transactionId, status, paidAt, refundStatus

VendorAssignment

id, bookingId, vendorJobId, vendorId, approvedCost, status, assignedAt

ExecutionJob

id, vendorAssignmentId, status, startTime, endTime

ExecutionTask

id, executionJobId, taskType, status, dueAt, completedAt, proofRequired

ExecutionProof

id, executionTaskId, assetId, proofType, uploadedBy, uploadedAt, verificationStatus

VisualAsset

id, sourceType, url, eventId, serviceId, vendorId, badge, metadata

ReferenceImage

id, eventId, eventServiceId, uploadedBy, assetId, analysisId

AIAnalysis

id, assetId, modelVersion, attributes, confidence, createdAt

EventState

id, eventId, version, stateJson, source, createdAt

17. EVENT STATE ENGINE

Event State is the single source of truth. Every meaningful mutation creates a versioned state transition. Use optimistic concurrency/version checks to prevent conflicting updates.

```
{
"eventType": "WEDDING",
"guestCount": 500,
"city": "Kolkata",
"budget": 500000,
"services": {
"decoration": {
"stage": {
"style": "elegant",
"primaryColor": "white",
"secondaryColor": "gold",
"floralDensity": "high",
"lighting": "warm",
"size": "10x20"
}
}
}
}
```

Sources

- CUSTOMER_MESSAGE
- BUTTON_SELECTION
- IMAGE_ANALYSIS
- AI_INFERENCE
- ADMIN
- VENDOR_RESPONSE
- VENDOR_PROPOSAL
- EXECUTION_UPDATE

Compatibility model

```
EventType
↓
ServiceCompatibility
↓
Service
↓
VendorCapability
↓
Eligible Opportunity
```

Example: Wedding → Mandap HIGH, Bridal Makeup HIGH, Corporate AV LOW, Exhibition Stall NONE. Corporate Conference → AV HIGH, Stage HIGH, Registration HIGH, Mandap NONE.

18. API + REAL-TIME CONTRACT

Auth

```
/api/auth
/api/customers
```

Events

```
/api/events
/api/events/:id/state
```

Catalogue & Visual

```
/api/services
/api/catalogue
/api/visuals
/api/references
/api/vision
```

Budget

```
/api/cost
/api/budget
/api/scenarios
```

Vendors

```
/api/vendors
/api/vendor-services
/api/vendor-availability
```

Matching

```
/api/matching
/api/opportunities
/api/vendor-jobs
/api/quotations
```

Commercial

```
/api/commercial
/api/bookings
/api/payments
```

Execution

```
/api/execution
/api/proofs
```

Real-time events

```
event.state.updated
visual.updated
budget.updated
vendor.response
vendor.scope.updated
quote.received
availability.updated
execution.updated
booking.updated
payment.updated
risk.updated
```

Recommended behavior: WebSocket/Socket.io events should be idempotent and include eventId, entityId, version, timestamp and actor/source. The backend remains authoritative; frontend treats real-time events as state updates, not as permission bypasses.

19. DATA ISOLATION & PERMISSIONS

Role	Can see	Must not see
Customer	Own event, configuration, customer price, visuals, availability result, booking, execution status	Vendor cost, margin, other vendors, internal score
Vendor	Opportunity scope, permitted references, AI brief, allowed target range, own quote, own jobs, earnings	Customer original budget, other vendor quotes, StarVnt margin, internal rules
Admin	Full operational and commercial view	—

Critical implementation requirement: enforce isolation in backend queries/DTOs/serializers and authorization policies. Do not merely hide sensitive fields in React.

20. STARVNT COMMAND CENTER

```
STARVNT COMMAND CENTER
■■■ Overview
■■■ Customers
■■■ Events
■■■ Aura+ Sessions
■■■ Live Events
■■■ Vendors
■■■ Vendor Network
■■■ Opportunities
■■■ Live Matching
■■■ Quotations
■■■ Bookings
■■■ Payments
■■■ Execution
■■■ Risks
■■■ Visual Catalogue
■■■ Portfolio
■■■ Pricing Rules
■■■ Commercial Rules
■■■ Service Catalogue
■■■ Event Templates
■■■ Vendor Performance
■■■ Analytics
```

Live Matching view

```
LIVE OPPORTUNITY
SV-1042
Premium Floral Stage
Kolkata · 500 Guests · 15 Dec
Search radius: 25 km
Eligible: 17
Responded: 6

A 12 sec ■42K
B 24 sec ■40K
C 41 sec ■43K
D 49 sec ■39K
E 53 sec ■44K
F 58 sec ■41K

Core Score
A 89
B 86
C 93
D 81
E 90
F 87
```

Admin sees the full picture. Customer sees only validated StarVnt-managed options. Vendor sees only its permitted opportunity information.

21. MATCHING ENGINE SPECIFICATION

Core matching is multi-factor. Do not implement “first responder wins”. Exact weights should be configurable by service and learned from historical performance, subject to business rules.

Candidate factors

- Scope Fit
- Availability
- Price Fit
- Distance
- Reliability
- Event-type Experience
- Style Match
- Capacity
- Response Speed
- Historical Performance

Illustrative scoring model (configurable, not hard-coded)

```
score =
w_scope * scopeFit
+ w_availability * availability
+ w_price * priceFit
+ w_distance * distanceFit
+ w_reliability * reliability
+ w_experience * eventExperience
+ w_style * styleMatch
+ w_capacity * capacityFit
+ w_response * responseSpeed
+ w_history * historicalPerformance

Hard constraints are evaluated BEFORE score ranking.
```

Hard constraints example

- Date unavailable → exclude.
- Service capability absent → exclude.
- Capacity insufficient → exclude.
- Compliance/verification failure → exclude.
- Scope cannot be executed → exclude or clarification state.
- Outside configured radius → exclude unless search wave expands.

22. SECURITY, SAFETY & RELIABILITY

• OTP/session security, rate limiting and device/session controls.

• JWT or secure HTTP-only cookie sessions depending on deployment architecture.

• Role-based authorization: CUSTOMER, VENDOR, ADMIN, OPERATIONS.

• Object-level authorization on every event/job/quote/payment endpoint.

• Signed or access-controlled media URLs for customer/vendor references and proof.

• Audit log for commercial changes, quote changes, scope changes, assignment, payment and admin overrides.

• Idempotency keys for booking/payment/quote submission.

• Database transactions for quote submission, assignment and commercial state changes.

• Vendor response timer must be server-authoritative; client countdown is visual only.

• Safety/compliance metadata for special effects, crowd control, security, medical and other regulated operational categories.

• AI outputs should be treated as suggestions/analysis unless promoted to approved structured state by the relevant core workflow.

23. RECOMMENDED TECHNICAL STACK

Layer	Recommended approach
Frontend Customer	Next.js + TypeScript + Tailwind CSS; responsive PWA/app shell; WebSocket client
Frontend Vendor	Next.js/React + TypeScript + Tailwind CSS; responsive dashboard/PWA
Backend Core	NestJS + TypeScript; REST + WebSocket
AI Service	Python FastAPI; vision analysis, extraction, orchestration helpers
Database	PostgreSQL + Prisma ORM
Realtime	Socket.io + Redis Pub/Sub
Media	Object storage such as S3-compatible storage; CDN; image processing
Payments	Razorpay or supported payment gateway
Deployment	Vercel for web; Render/Railway or equivalent for backend; managed PostgreSQL
Observability	Structured logs, metrics, traces, error tracking and audit events

Architecture principle: keep AI services separate from authoritative transactional logic. Aura+ may orchestrate and explain, but Core owns state, constraints, pricing, matching, booking and execution transitions.

24. SUGGESTED MONOREPO STRUCTURE

```
starvnt-event-os/
■■■ apps/
■ ■■■ customer-web/
■ ■■■ vendor-web/
■ ■■■ admin-web/
■ ■■■ mobile/ # later
■
■■■ services/
■ ■■■ api-gateway/
■ ■■■ aura-ai/
■ ■■■ matching-engine/
■ ■■■ cost-engine/
■ ■■■ visual-engine/
■ ■■■ realtime-hub/
■ ■■■ media-engine/
■ ■■■ notification-service/
■
■■■ packages/
■ ■■■ ui/
■ ■■■ event-schema/
■ ■■■ service-catalogue/
■ ■■■ vendor-capabilities/
■ ■■■ auth/
■ ■■■ realtime/
■ ■■■ shared-types/
■
■■■ prisma/
■ ■■■ schema.prisma
■ ■■■ migrations/
■
■■■ docs/
■■■ product/
■■■ api/
■■■ event-templates/
■■■ service-catalogue/
■■■ vendor-playbooks/
```

Suggested backend domain modules

```
auth
customers
events
event-state
catalogue
visuals
vision
budget
cost
vendors
availability
matching
opportunities
vendor-jobs
quotations
commercial
bookings
payments
execution
proofs
notifications
analytics
audit
```

25. END-TO-END DEVELOPER FLOWS

A. Customer creates a service

```
Customer clicks Add
→ POST /events/:id/services
→ Event State transaction
→ Cost recalculation
→ Visual update event
→ If actionable: matching job created
→ Vendor opportunity broadcast
→ Responses collected
→ Core validation
→ Customer availability state updated
```

B. Customer uploads reference

```
Upload media
→ create ReferenceImage
→ Vision service analyzes
→ AIAnalysis saved
→ Aura+ converts analysis to candidate ServiceConfiguration
→ customer confirms / edits
→ Event State version increments
→ Visual + Cost + Matching recalculate
```

C. Vendor responds

```
Vendor opens opportunity
→ Available
→ server records response timestamp
→ scope confirmation
→ quote draft
→ vendor submits
→ quote validated
→ Core scoring
→ eligible result stored
→ customer-facing availability updated
```

D. Customer changes scope after quote

```
Customer edit
→ Event State version increments
→ detect affected service
→ invalidate/review previous quote if necessary
→ recalculate price
→ send vendor scope-change event
→ vendor Accept / Requote / Cannot Execute
→ Core validates
→ customer sees updated service state
```

26. EVENT DIGITAL TWIN

The Event Digital Twin is the operational representation shared across customer, vendor and StarVnt Core. Each role sees a permissioned projection of the same underlying state.

```
EVENT DIGITAL TWIN
■
■■■ WHO
■ Guests / Hosts / Vendors / Staff
■
■■■ WHAT
■ Services / Requirements / Assets
■
■■■ WHERE
■ Venue / Location / Zones
■
■■■ WHEN
■ Timeline / Milestones / Deadlines
■
■■■ HOW MUCH
■ Customer Price / Vendor Cost / Margin (permissioned)
■
■■■ WHO EXECUTES
■ Vendor Assignments
■
■■■ WHAT IS HAPPENING
■ Live Execution State
■
■■■ WHAT IS AT RISK
Risk / Exception State
```

Customer projection: experience/readiness. Vendor projection: job/scope/execution. Admin projection: full commercial + operational state.

27. VENDOR PERFORMANCE + LEARNING LOOP

After each event, store actual operational outcomes. This data improves future matching and pricing rules.

- Quoted amount vs approved amount vs actual cost where available.
- Response time.
- Availability accuracy.
- Setup delay.
- Scope variance.
- Cancellation/no-show.
- Customer feedback.
- Quality issue.
- Proof completeness.
- On-time completion.
- Event-type experience.
- Service-specific reliability.

```
Completed Job
↓
Actuals + Feedback
↓
Vendor Performance
↓
Matching Features
↓
Future Vendor Eligibility / Ranking
```

Important: performance metrics should be transparent internally, versioned, and protected from arbitrary manipulation.

28. EVENT TEMPLATE ENGINE

Templates are configuration, not hard-coded frontend flows. A template defines possible modules, dependencies, default priorities, compatibility and question candidates. Aura+ chooses only the next relevant decision.

```
Event Template
■■■ Event Type
■■■ Modules
■■■ Service Compatibility
■■■ Question Candidates
■■■ Skip Rules
■■■ Required Conditions
■■■ Visual Categories
■■■ Budget Rules
■■■ Vendor Categories
■■■ Execution Milestones
■■■ Risk Rules
```

Example: Bengali Wedding Template

Identity • Venue • Guests • Budget • Food • Decoration • Mandap • Jaimala • Photography • Videography • Makeup • Mehendi • Entertainment • Bride/Groom • Transport • Guest Stay • Invitation • Gifting • Timeline • Special Requirements

Example: 500-person Corporate Conference Template

Identity • Venue • Delegates • Agenda • Stage • AV • LED • Branding • Registration • Speaker • Catering • Photography • Streaming • Transport • Accommodation • Security • Event Staffing • Timeline • Technical Rehearsal

29. QA / ACCEPTANCE CRITERIA

1. Customer can create an event without completing every module.
2. Customer can add a service without being forced through Aura+ questions.
3. Customer can upload a reference image and receive structured visual analysis.
4. Changing a service configuration updates Event State, visual state and budget.
5. Removing a service removes or invalidates dependent vendor requirements where appropriate.
6. Vendor never receives customer original budget when budget isolation is enabled.
7. Vendor response timer is server-authoritative and expires correctly.
8. Fastest responder is not automatically selected.
9. Unavailable vendors cannot be presented as available.
10. AI concept is clearly labeled and never silently becomes bookable.
11. Customer sees customer price only.
12. Vendor sees only permitted job/commercial data.
13. Admin can trace the full lifecycle with audit events.
14. Quote and booking operations are idempotent.
15. Scope changes propagate to affected vendors and invalidate stale quotes when required.
16. Execution milestones can be updated with proof.
17. Real-time events recover correctly after reconnect.
18. New event types can be added through configuration without rewriting the core event engine.

30. IMPLEMENTATION PHASES

Phase 1 — Foundation

Auth, Customer, Vendor, Event State, Service Catalogue, Event Templates, basic Admin.

Phase 2 — Aura+ Customer

Conversation engine, direct Add/Edit, visual cards, reference upload, Vision Analysis.

Phase 3 — Visual + Cost

Visual Engine, Live Canvas, Budget Engine, What-if Engine.

Phase 4 — Vendor Network

Vendor onboarding, capability graph, availability, portfolio, opportunities.

Phase 5 — Matching

Geo Engine, eligibility, 60-sec response, scoring, scope validation.

Phase 6 — Quotation + Commercial

Vendor quote builder, commercial isolation, customer price, approval.

Phase 7 — Booking + Payment

Booking lifecycle, payment gateway, assignment.

Phase 8 — Execution

Vendor execution, proof, customer tracking, Digital Twin.

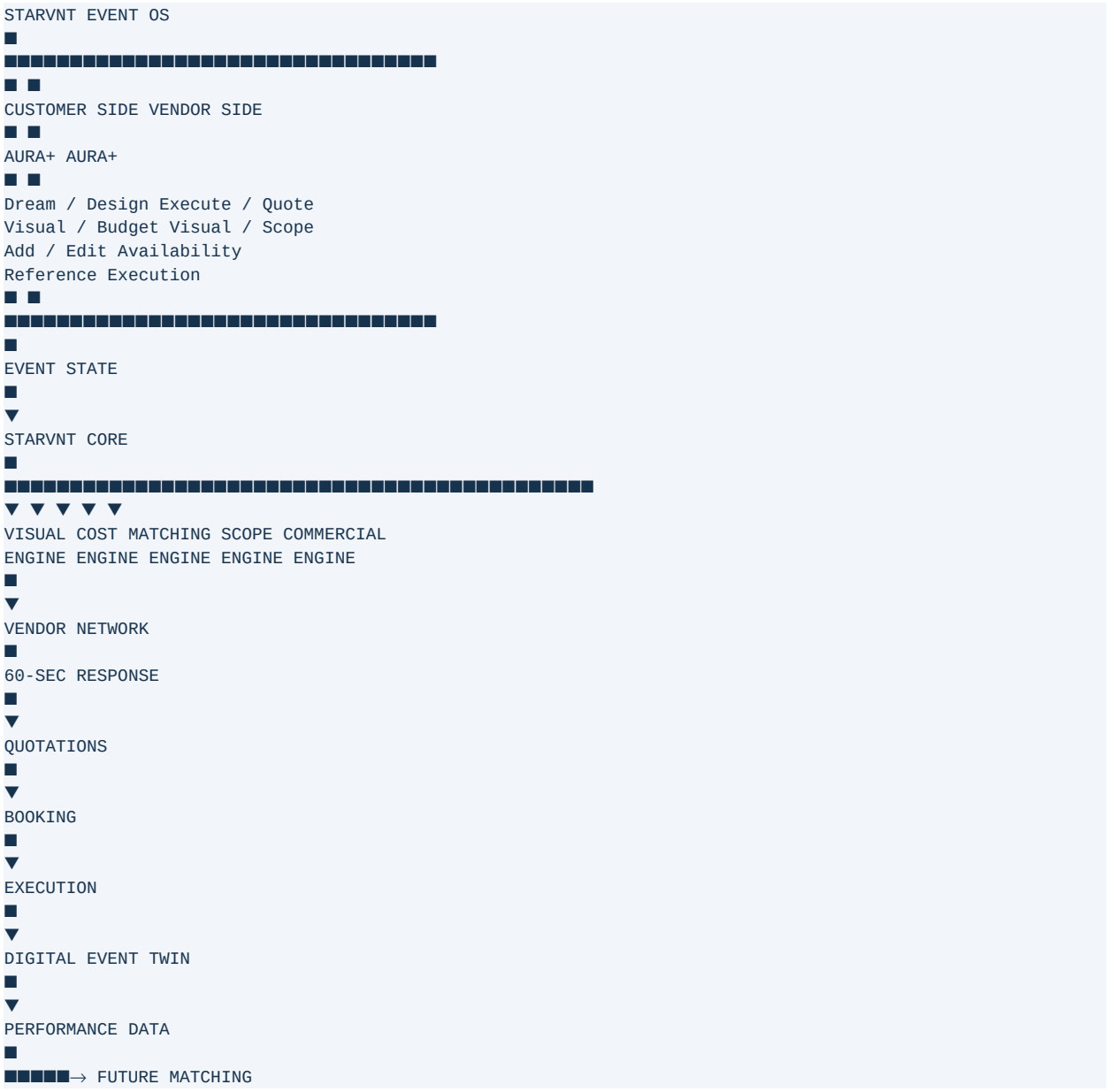
Phase 9 — Learning

Performance metrics, historical matching, risk engine, analytics.

Phase 10 — Expansion

More event templates, more vendor categories, mobile apps, advanced AI and automation.

31. FINAL PRODUCT MODEL



Final experience

Customer designs. Aura+ understands. StarVnt Core orchestrates. Vendor executes. Aura+ keeps both sides synchronized.

Final product definition

Any Event → Any Service → Any Vendor → One Intelligent Event OS.

Developer rule: build the platform as a configuration-driven Event OS. Do not hard-code a wedding-only journey. Event Type Engine + Service Catalogue + Vendor Capability Graph + Event State Engine are the extensibility foundation.